

[< Back](#)

RESEARCH ARTICLE

Toward sustainable corporate behavior: The effect of the critical mass of female directors on environmental, social, and governance disclosure

Sara De Masi , Agnieszka Słomka-Gołębiowska, Claudio Becagli, Andrea Paci

First published: 04 January 2021

<https://doi.org/10.1002/bse.2721>[↗ VIEW METRICS](#)Accessibility issue? [Request accessibility update.](#)

PDF

Abstract

Boards of directors have recently become more attentive to their stakeholders' concerns, providing more transparent information and adopting more sustainable business strategies. This study investigates the influence of a critical mass of women on boards on the environmental, social, and governance (ESG) disclosure score and its three components separately. Using a sample of the FTSE-MIB listed companies in the 2005–2017 period, we show that reaching a critical mass of female board members—going from one or two women to at least three—enhances the level of ESG disclosure. The results also show that the critical mass of female board members has a positive influence on every component of the ESG score, with the highest contribution of women reaching the governance score. These findings provide insights to shareholders and policymakers and suggest that a critical mass of female board members is particularly effective in improving transparency, and it can be seen as a mechanism to transit to stakeholder governance, fostering more sustainable behavior in firms.

REFERENCES

Adams, R. B., & Ferreira, D. (2009). Women in the boardroom and their impact on governance and performance. *Journal of Financial Economics*, 94(2), 291–309. <https://doi.org/10.1016/j.jfineco.2008.10.007>

[< Back](#)

Alkalbani, N., Cuomo, F., & Mallin, C. (2019). Gender diversity and say-on-pay: Evidence from UK remuneration committees. *Corporate Governance: An International Review*, 27, 378–400. <https://doi.org/10.1111/corg.12292>

[Web of Science®](#)[Google Scholar](#)

Allen, M. W., & Craig, C. A. (2016). Rethinking corporate social responsibility in the age of climate change: A communication perspective. *International Journal of Corporate Social Responsibility*, 1(1), 1–11. <https://doi.org/10.1186/s40991-016-0002-8>

[Google Scholar](#)

Armstrong, C. S., Core, J. E., & Guay, W. R. (2014). Do Independent Directors Cause Improvements in Firm Transparency? *Journal of Financial Economics*, 113(3), 383–403. <https://doi.org/10.1016/j.jfineco.2014.05.009>

[Web of Science®](#)[Google Scholar](#)

Atif, M., Alam, S., & Hossain, M. (2020). Firm sustainable investment: Are female directors greener? *Business Strategy and the Environment*, 29, 1–21. <https://doi.org/10.1002/bse.2588>

[Web of Science®](#)[Google Scholar](#)

Aureli, S., Del Baldo, M., Lombardi, R., & Nappo, F. (2020). Nonfinancial reporting regulation and challenges in sustainability disclosure and corporate governance practices. *Business Strategy and the Environment*, 29, 2392–2403. <https://doi.org/10.1002/bse.2509>

[Web of Science®](#)[Google Scholar](#)

Aureli, S., Gigli, S., Medei, R., & Supino, E. (2020). The value relevance of environmental, social, and governance disclosure: Evidence from Dow Jones Sustainability World Index listed companies. *Corporate Social Responsibility and Environmental Management*, 27(1), 43–52. <https://doi.org/10.1002/csr.1772>

[Web of Science®](#)[Google Scholar](#)

[Back](#)

369–383. <https://doi.org/10.1007/s10551-015-2759-1>

[Web of Science®](#)

[Google Scholar](#)

Ben-Amar, W., & McIlkenny, P. (2015). Board effectiveness and the voluntary disclosure of climate change information. *Business Strategy and the Environment*, 24, 704–719. <https://doi.org/10.1002/bse.1840>

[Web of Science®](#)

[Google Scholar](#)

Bernardi, R., Bosco, S., & Vassill, K. (2006). Does female representation on board of directors associates with Fortune's 100 Best Companies to Work For list? *Business and Society*, 45, 235–248. <https://doi.org/10.1177/0007650305283332>

[Google Scholar](#)

Bernardi, R. A., Bosco, S., & Columb, V. L. (2009). Does female representation on boards of directors associate with the 'Most ethical companies' list? *Corporate Reputation Review*, 12(3), 270–280. <https://doi.org/10.1057/crr.2009.15>

[Google Scholar](#)

Birindelli, G., Iannuzzi, A. P., & Savioli, M. (2019). The impact of women leaders on environmental performance: Evidence on gender diversity in banks. *Corporate Social Responsibility and Environmental Management*, 26, 1485–1499. <https://doi.org/10.1002/csr.1762>

[Web of Science®](#)

[Google Scholar](#)

Bloomberg, L. P. (2014). *Look Beyond: Bloomberg for environmental, social and governance analysis*.

[Google Scholar](#)

Bloomberg, L. P. (2017). ESG disclosure scores. *Research domain, Software and database*, New York.

[Google Scholar](#)

[Back](#)[Web of Science®](#)[Google Scholar](#)

Bonacchi, M., & Rinaldi, L. (2007). DartBoards and clovers as new tools in sustainability planning and control. *Business Strategy and the Environment*, 16, 461–473. <https://doi.org/10.1002/bse.596>

[Google Scholar](#)

Bravo, F., & Reguera-Alvarado, N. (2018). Sustainable development disclosure: Environmental, social, and governance reporting and gender diversity in the audit committee. *Business Strategy and the Environment*, 28, 418–429. <https://doi.org/10.1002/bse.2258>

[Web of Science®](#)[Google Scholar](#)

Bugeja, M., Matolcsy, Z., & Spiropoulos, H. (2016). The association between gender-diverse compensation committees and CEO compensation. *Journal of Business Ethics*, 139(2), 375–390. <https://doi.org/10.1007/s10551-015-2660-y>

[Web of Science®](#)[Google Scholar](#)

Byron, K., & Post, C. (2016). Women on boards of directors and corporate social performance: A meta-analysis. *Corporate Governance: An International Review*, 24(4), 428–442. <https://doi.org/10.1111/corg.12165>

[Web of Science®](#)[Google Scholar](#)

Cardoni, A., Kiseleva, E., & Lombardi, R. (2020). A sustainable governance model to prevent corporate corruption: Integrating anticorruption practices, corporate strategy and business processes. *Business Strategy and the Environment*, 29, 1173–1185. <https://doi.org/10.1002/bse.2424>

[Web of Science®](#)[Google Scholar](#)

Chen, J., Leung, W. S., & Goergen, M. (2017). The impact of board gender composition on dividend payouts. *Journal of Corporate Finance*, 43, 86–105. <https://doi.org/10.1016/j.jcorpfin.2017.01.001>

[< Back](#)

Cordazzo, M., Bini, L., & Marzo, G. (2020). Does the EU directive on non-financial information influence the value relevance of ESG disclosure? Italian evidence. *Business Strategy and the Environment*, 29, 1–14. <https://doi.org/10.1002/bse.2589>

[Web of Science®](#)[Google Scholar](#)

Cordeiro, J. J., Profumo, G., & Tutore, I. (2020). Board gender diversity and corporate environmental performance: The moderating role of family and dual-class majority ownership structures. *Business Strategy and the Environment*, 29, 1127–1144. <https://doi.org/10.1002/bse.2421>

[Web of Science®](#)[Google Scholar](#)

Cucari, N., Esposito De Falco, S., & Orlando, B. (2018). Diversity of board of directors and environmental social governance: Evidence from Italian listed companies. *Corporate Social Responsibility and Environmental Management*, 25, 250–266. <https://doi.org/10.1002/csr.1452>

[Web of Science®](#)[Google Scholar](#)

Cumming, D., Leung, T. Y., & Rui, O. (2015). Gender diversity and securities fraud. *Academy of Management Journal*, 58(5), 1572–1593. <https://doi.org/10.5465/amj.2013.0750>

[Web of Science®](#)[Google Scholar](#)

Cunha, F. A. F., De Oliveira, E. M., Orsato, R. J., Klotzle, M. C., Cyrino Oliveira, F. L., & Caiado, R. G. G. (2020). Can sustainable investments outperform traditional benchmarks? Evidence from global stock markets. *Business Strategy and the Environment*, 29, 682–697. <https://doi.org/10.1002/bse.2397>

[Web of Science®](#)[Google Scholar](#)

Dahlsrud, A. (2008). How corporate social responsibility is defined: An analysis of 37 definitions. *Corporate Social Responsibility and Environmental Management*, 15, 1–13. <https://doi.org/10.1002/csr.132>

[Web of Science®](#)[Google Scholar](#)

[< Back](#)

3916596283003

[Web of Science®](#) [↗](#)[Google Scholar](#) [↗](#)

De Masi, S., Słomka-Gołębiowska, A., & Paci, A. (2019). Women on board, quota and board tasks: The sanctions matter. Euram Conference Proceeding.

[Google Scholar](#) [↗](#)

Eagly, A. H., Johannesen-Schmidt, M. C., & Van Engen, M. L. (2003). Transformational, transactional, and laissez-faire leadership styles: A meta-analysis comparing women and men. *Psychological Bulletin*, 129, 569–591. <https://doi.org/10.1037/0033-2909.129.4.569>

[PubMed](#) [↗](#)[Web of Science®](#) [↗](#)[Google Scholar](#) [↗](#)

Eagly, A. H., Karau, S. J., & Makhijani, M. G. (1995). Gender and the effectiveness of leaders: A meta-analysis. *Psychological Bulletin*, 117, 125–145. <https://doi.org/10.1037/0033-2909.117.1.125>

[CAS](#) [↗](#)[PubMed](#) [↗](#)[Web of Science®](#) [↗](#)[Google Scholar](#) [↗](#)

Eagly, A. H., & Wood, W. (2013). The nature–nurture debates: 25 years of challenges in understanding the psychology of gender. *Perspectives on Psychological Science*, 8(3), 340–357. <https://doi.org/10.1177/1745691613484767>

[PubMed](#) [↗](#)[Web of Science®](#) [↗](#)[Google Scholar](#) [↗](#)

Eccles, R. G., Serafeim, G., & Krzus, M. P. (2011). Market interest in nonfinancial information. *Journal of Applied Corporate Finance*, 23(4), 113–127. <https://doi.org/10.1111/j.1745-6622.2011.00357>

[Google Scholar](#) [↗](#)

Elmagrhi, M. H., Ntim, C. G., Elamer, A. A., & Zhang, Q. (2019). A study of environmental policies and regulations, governance structures, and environmental performance: The role of female directors. *Business Strategy and the Environment*, 28, 206–220. <https://doi.org/10.1002/bse.2250>

[< Back](#)

Ely, R., & Padavic, I. (2007). A feminist analysis of organizational research on sex differences. *Academy of Management Review*, 32(4), 1121–1143. <https://doi.org/10.5465/amr.2007.26585842>

[Web of Science®](#) [Google Scholar](#)

Fernandez-Sanchez, J. L., Luna, L., & Baraibar-Diez, E. (2011). The relationship between corporate governance and corporate social behavior: A structural equation model analysis. *Corporate Social Responsibility and Environmental Management*, 18(2), 91–101. <https://doi.org/10.1002/csr.244>

[Web of Science®](#) [Google Scholar](#)

Franke, G. R., Crown, D. F., & Spake, D. F. (1997). Gender differences in ethical perceptions of business practices: A social role theory perspective. *Journal of Applied Psychology*, 82(6), 920–934. <https://doi.org/10.1037/0021-9010.82.6.920>

[CAS](#) [PubMed](#) [Web of Science®](#) [Google Scholar](#)

Galbreath, J. (2013). ESG in focus: The Australian evidence. *Journal of Business Ethics*, 118(3), 529–541. <https://doi.org/10.1007/s10551-012-1607-9>

[Web of Science®](#) [Google Scholar](#)

Galbreath, J. (2018). Do boards of directors influence corporate sustainable development? An attention-based analysis. *Business Strategy and the Environment*, 27, 742–756. <https://doi.org/10.1002/bse.2028>

[Web of Science®](#) [Google Scholar](#)

Glass, C., Cook, A., & Ingersoll, A. R. (2016). Do women leaders promote sustainability? Analyzing the effect of corporate governance composition on environmental performance. *Business Strategy and the Environment*, 25(7), 495–511. <https://doi.org/10.1002/bse.1879>

[Web of Science®](#) [Google Scholar](#)

[< Back](#)[Web of Science®](#)[Google Scholar](#)

Hambrick, D. C., Cho, T. S., & Chen, M. J. (1996). The influence of top management team heterogeneity on firms' competitive moves. *Administrative Science Quarterly*, 41, 659–684. <https://doi.org/10.2307/2393871>

[Web of Science®](#)[Google Scholar](#)

Hambrick, D. C., & Mason, P. A. (1984). Upper echelons: The organization as a reflection of its top managers. *Academy of Management Review*, 9(2), 193–206. <https://doi.org/10.5465/amr.1984.4277628>

[Web of Science®](#)[Google Scholar](#)

Harrigan, K. R. (1981). Numbers and positions of women elected to corporate boards. *Academy of Management Journal*, 24, 619–625.

[Web of Science®](#)[Google Scholar](#)

Helfaya, A., & Moussa, T. (2017). Do board's corporate social responsibility strategy and orientation influence environmental sustainability disclosure? UK evidence. *Business Strategy and the Environment*, 26, 1061–1077. <https://doi.org/10.1002/bse.1960>

[Web of Science®](#)[Google Scholar](#)

Hillman, A. J., Cannella, A. A., & Harris, I. C. (2002). Women and racial minorities in the boardroom: How do directors differ? *Journal of Management*, 28(6), 747–763. <https://doi.org/10.1177/014920630202800603>

[Web of Science®](#)[Google Scholar](#)

Hoang, T.-H., Przychodzen, W., Przychodzen, J., & Segbotangni, E. A. (2020). Does it pay to be green? A disaggregated analysis of U.S. firms with green patents. *Business Strategy and the Environment*, 29, 1331–1361. <https://doi.org/10.1002/bse.2437>

[< Back](#)

Hsueh, L. (2019). Opening up the firm: What explains participation and effort in voluntary carbon disclosure by global businesses? An analysis of internal firm factors and dynamics. *Business Strategy and the Environment*, 28, 1302–1322. <https://doi.org/10.1002/bse.2317>

[Web of Science®](#)[Google Scholar](#)

Hur, W. M., Kim, H., & Jang, J. H. (2016). The role of gender differences in the impact of CSR perceptions on corporate marketing outcomes. *Corporate Social Responsibility and Environmental Management*, 23, 345–357. <https://doi.org/10.1002/csr.1380>

[Web of Science®](#)[Google Scholar](#)

Husted, N. W., & de Sousa-Filho, J. M. (2019). Board structure and environmental, social, and governance disclosure in Latin America. *Journal of Business Research*, 102, 220–227. <https://doi.org/10.1016/j.jbusres.2018.01.017>

[Web of Science®](#)[Google Scholar](#)

Jaffee, S., & Hyde, J. S. (2000). Gender differences in moral orientation: A meta-analysis. *Psychological Bulletin*, 126(5), 703–726. <https://doi.org/10.1037/0033-2909.126.5.703>

[CAS](#)[PubMed](#)[Web of Science®](#)[Google Scholar](#)

Jizi, M. (2017). The influence of board composition on sustainable development disclosure. *Business Strategy and the Environment*, 26, 640–655. <https://doi.org/10.1002/bse.1943>

[Web of Science®](#)[Google Scholar](#)

Joecks, J., Pull, K., & Vetter, K. (2013). Gender diversity in the boardroom and firm performance: What exactly constitutes a “critical mass?”. *Journal of Business Ethics*, 118(1), 61–72. <https://doi.org/10.1007/s10551-012-1553-6>

[Web of Science®](#)[Google Scholar](#)

[< Back](#)

International research and practice (pp. 17–31). Cheltenham: Edward Elgar.

[Web of Science®](#)[Google Scholar](#)

Kanter, R. M. (1977). *Men and women of the corporation*. New York: Basic Books.

[Google Scholar](#)

Kirsch, A. (2017). The gender composition of corporate boards: A review and research agenda. *The Leadership Quarterly*, 29(2), 346–364.

[Web of Science®](#)[Google Scholar](#)

Konrad, A. M., Kramer, V., & Erkut, S. (2008). The impact of three or more women on corporate boards. *Organizational Dynamics*, 37(2), 145–164. <https://doi.org/10.1016/j.orgdyn.2008.02.005>

[Web of Science®](#)[Google Scholar](#)

Lagasio, V., & Cucari, N. (2019). Corporate governance and environmental social governance disclosure: A meta-analytical review. *Corporate Social Responsibility and Environmental Management*, 26, 701–711. <https://doi.org/10.1002/csr.1716>

[Web of Science®](#)[Google Scholar](#)

Lai, A., Melloni, G., & Stacchezzini, R. (2016). Corporate sustainable development: Is ‘integrated reporting’ a legitimization strategy? *Business Strategy and the Environment*, 25(3), 165–177. <https://doi.org/10.1002/bse.1863>

[Web of Science®](#)[Google Scholar](#)

Li, J., Zhao, F., Chen, S., Jiang, W., Liu, T., & Shi, S. (2017). Gender diversity on boards and firms' environmental policy. *Business Strategy and the Environment*, 26, 306–315. <https://doi.org/10.1002/bse.1918>

[Web of Science®](#)[Google Scholar](#)

[Back](#)[Web of Science®](#)[Google Scholar](#)

Lu, J., & Herremans, I. M. (2019). Board gender diversity and environmental performance: An industries perspective. *Business Strategy and the Environment*, 28, 1449–1464. <https://doi.org/10.1002/bse.2326>

[Web of Science®](#)[Google Scholar](#)

Lucas-Pérez, M. E., Mínguez-Vera, A., Baixauli-Soler, J. S., Martín-Ugedo, J. F., & Sánchez-Marín, G. (2015). Women on the board and managers' pay: Evidence from Spain. *Journal of Business Ethics*, 129(2), 265–280. <https://doi.org/10.1007/s10551-014-2148-1>

[Web of Science®](#)[Google Scholar](#)

Machold, S., Huse, M., Hansen, K., & Brogi, M. (2013). *Getting women on to corporate boards: A snowball starting in Norway*. Cheltenham: Edward Elgar Publishing.

[Google Scholar](#)

Manita, R., Bruna, M. G., Dang, R., & Houanti, L. (2018). Board gender diversity and ESG disclosure: Evidence from the USA. *Journal of Applied Accounting Research*, 19(2), 206–224. <https://doi.org/10.1108/JAAR-01-2017-0024>

[Web of Science®](#)[Google Scholar](#)

Michelon, G., & Parbonetti, A. (2012). The effect of corporate governance on sustainability disclosure. *Journal of Management and Governance*, 16(3), 477–509. <https://doi.org/10.1007/s10997-010-9160-3>

[Web of Science®](#)[Google Scholar](#)

Mio, C., Venturelli, A., & Leopizzi, R. (2015). Management by objectives and corporate social responsibility disclosure. *Accounting, Auditing & Accountability Journal*, 28, 325–364. <https://doi.org/10.1108/AAAJ-09-2013-1480>

[Web of Science®](#)[Google Scholar](#)

[Back](#)

1404. <https://doi.org/10.3390/su11051404>

[Web of Science®](#)

[Google Scholar](#)

Mize, T. D., Doan, L., & Long, J. S. (2019). A general framework for comparing predictions and marginal effects across models. *Sociological Methodology*, 49(1), 152–189. <https://doi.org/10.1177/0081175019852763>

[Web of Science®](#)

[Google Scholar](#)

Nadeem, M., Bahadar, S., Gull, A. A., & Iqbal, U. (2020). Are women eco-friendly? Board gender diversity and environmental innovation. *Business Strategy and the Environment*, 29, 1–16. <https://doi.org/10.1002/bse.2563>

[Web of Science®](#)

[Google Scholar](#)

Nalikka, A. (2009). Impact of gender diversity on the extent of voluntary disclosure in annual reports. *Accounting and Taxation*, 1(1), 101–113.

[Google Scholar](#)

Orazalin, N. (2020). Do board sustainability committees contribute to corporate environmental and social performance? The mediating role of corporate social responsibility strategy. *Business Strategy and the Environment*, 29, 140–153. <https://doi.org/10.1002/bse.2354>

[Web of Science®](#)

[Google Scholar](#)

Ortiz-de-Mandojana, N., & Aragon-Correa, J. A. (2015). Boards and sustainability: The contingent influence of director interlocks on corporate environmental performance. *Business Strategy and the Environment*, 24, 499–517. <https://doi.org/10.1002/bse.1833>

[Web of Science®](#)

[Google Scholar](#)

Pan, C., Guo, H., Jiang, Y., Wang, H., & Qi, W. (2020). The double effects of female executives' participation on corporate sustainable competitive advantage through unethical environmental

[< Back](#)[Web of Science®](#)[Google Scholar](#)

Pan, Y., & Sparks, J. R. (2012). Predictors, consequence, and measurement of ethical judgments: Review and meta-analysis. *Journal of Business Research*, 65(1), 84–91. <https://doi.org/10.1016/j.jbusres.2011.02.002>

[Web of Science®](#)[Google Scholar](#)

Powell, G. N. (1990). One more time: Do female and male managers differ? *The Executive*, 4, 68–75.

[Google Scholar](#)

Qureshi, M. A., Kirkerud, S., Theresa, K., & Ahsan, T. (2020). The impact of sustainability (environmental, social, and governance) disclosure and board diversity on firm value: The moderating role of industry sensitivity. *Business Strategy and the Environment*, 29, 1199–1214. <https://doi.org/10.1002/bse.2427>

[Web of Science®](#)[Google Scholar](#)

Rao, K., & Tilt, C. (2016). Board composition and corporate social responsibility: The role of diversity, gender, strategy and decision making. *Journal of Business Ethics*, 138(2), 327–347. <https://doi.org/10.1007/s10551-015-2613-5>

[Web of Science®](#)[Google Scholar](#)

Rupley, K. H., Brown, D., & Marshall, R. S. (2012). Governance, media and the quality of environmental disclosure. *Journal of Accounting and Public Policy*, 31(6), 610–640. <https://doi.org/10.1016/j.jaccpubpol.2012.09.002>

[Web of Science®](#)[Google Scholar](#)

Sassen, R., Hinze, A. K., & Hardeck, I. (2016). Impact of ESG factors on firm risk in Europe. *Journal of Business Economics*, 86(8), 867–904. <https://doi.org/10.1007/s11573-016-0819-3>

[Google Scholar](#)

[Back](#)

07.10.002

[Google Scholar](#) 

Terjesen, S., Sealy, R., & Singh, V. (2009). Women directors on corporate boards: A review and research agenda. *Corporate Governance: An International Journal*, 17, 320–337. <https://doi.org/10.1111/j.1467-8683.2009.00742.x>

[Web of Science®](#) [Google Scholar](#) 

Tingbani, I., Chithambo, L., Tauringana, V., & Papanikolaou, N. (2020). Board gender diversity, environmental committee and greenhouse gas voluntary disclosures. *Business Strategy and the Environment*, 29, 1–17. <https://doi.org/10.1002/bse.2495>

[Web of Science®](#) [Google Scholar](#) 

Torchia, M., Calabrò, A., & Huse, M. (2011). Women directors on corporate boards: From tokenism to critical mass. *Journal of Business Ethics*, 102, 299–317. <https://doi.org/10.1007/s10551-011-0815-z>

[Web of Science®](#) [Google Scholar](#) 

Torchia, M., Seierstad, C., Warner-Søderholm, G., & Huse, M. (2017). Women on boards: Beyond the institutional setting—The role of stakeholders and actors. *Journal of Business Ethics*, 141(2), 289–315.

[Google Scholar](#) 

Upadhyay, A., & Zeng, H. (2014). Gender and ethnic diversity on boards and corporate information environment. *Journal of Business Research*, 67(11), 2456–2463. <https://doi.org/10.1016/j.jbusres.2014.03.005>

[Web of Science®](#) [Google Scholar](#) 

US Business Roundtable (2019). *Business Roundtable Redefines the Purpose of a Corporation to Promote 'An Economy That Serves All Americans'*. <https://www.businessroundtable.org/business-roundtable-redefines-the-purpose-of-a-corporation-to-promote-an-economy-that-serves-all-americans>

[< Back](#)

Usman, M., Zhang, J., Wang, F., Sun, J., & Makki, M. A. M. (2018). Gender diversity in compensation committees and CEO pay: Evidence from China. *Management Decision*, 56(5), 1065–1087. <https://doi.org/10.1108/MD-09-2017-0815>

[Web of Science®](#)[Google Scholar](#)

Verbeek, M. (2008). *A guide to modern econometrics*. Chichester, UK: John Wiley & Sons, Ltd.

[Google Scholar](#)

Wang, J., & Coffey, B. S. (1992). Board composition and corporate philanthropy. *Journal of Business Ethics*, 11, 771–778. <https://doi.org/10.1007/BF00872309>

[Web of Science®](#)[Google Scholar](#)

Weesie, J. (1999). Seemingly unrelated estimation and cluster-adjusted sandwich estimator. *Stata Technical Bulletin*, 9, 231–248.

[Google Scholar](#)

Westphal, J., & Milton, L. (2000). How experience and network ties affect the influence of demographic minorities on corporate boards. *Administrative Science Quarterly*, 45(2), 366–398. <https://doi.org/10.2307/2667075>

[Web of Science®](#)[Google Scholar](#)

Widyawati, L. (2020). A systematic literature review of socially responsible investment and environmental social governance metrics. *Business Strategy and the Environment*, 29, 619–637. <https://doi.org/10.1002/bse.2393>

[Web of Science®](#)[Google Scholar](#)

Williams, R. (2003). Women on corporate boards of directors and their influence on corporate philanthropy. *Journal of Business Ethics*, 42, 1–10. <https://doi.org/10.1023/A:1021626024014>

[< Back](#)

World Economic Forum (2020). *Why we need the 'Davos Manifesto' for a better kind of capitalism*, <https://www.weforum.org/agenda/2019/12/why-we-need-the-davos-manifesto-for-better-kind-of-capitalism/>

[Google Scholar](#) 

Yu, E. P., Guo, C. Q., & Luu, B. V. (2018). Environmental, social and governance transparency and firm value. *Business Strategy and the Environment*, 27, 987–1004. <https://doi.org/10.1002/bse.2047>

[Web of Science®](#)  [Google Scholar](#) 

Zanoni, P., Janssens, M., Benschop, Y., & Nkomo, S. (2010). Guest editorial: Unpacking diversity, grasping inequality: Rethinking difference through critical perspectives. *Organization*, 17(1), 9–29. <http://doi.org/10.1177/1350508409350344>

[Web of Science®](#)  [Google Scholar](#) 

Zelechowski, D. D., & Bilimoria, D. (2004). Characteristics of women and men corporate inside directors in the US. *Corporate Governance: An International Review*, 12(3), 337–342. <https://doi.org/10.1111/j.1467-8683.2004.00374.x>

[Web of Science®](#)  [Google Scholar](#) 

Citing Literature



[Download PDF](#)

ABOUT WILEY ONLINE LIBRARY

[Privacy Policy](#)

[Terms of Use](#)

[< Back](#)[Accessibility](#)[Wiley Research DE&I Statement and Publishing Policies](#)

HELP & SUPPORT

[Contact Us](#)[Training and Support](#)[DMCA & Reporting Piracy](#)[Sitemap](#)

OPPORTUNITIES

[Subscription Agents](#)[Advertisers & Corporate Partners](#)

CONNECT WITH WILEY

[The Wiley Network](#)[Wiley Press Room](#)

Copyright © 1999-2026 John Wiley & Sons, Inc or related companies. All rights reserved, including rights for text and data mining and training of artificial intelligence technologies or similar technologies.